

SKILL - 21:

Code:

```
GNU nano 7.2 calculator.h
#ifndef CALCULATOR_H
#define CALCULATOR_H

int calculate(const char *operation, double a, double b, double *result);

#endif

[ Read 6 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_/ Go To Line ^-E Redo      ^-G Copy
```

```
GNU nano 7.2 logger.c
#include <stdio.h>
#include <time.h>
#include "logger.h"

void log_error(const char *message)
{
    FILE *file = fopen("error.log", "a");

    if (file == NULL) {
        perror("Unable to open error log");
        return;
    }

    time_t now = time(NULL);
    struct tm *time_info = localtime(&now);
    char timestamp[32] = "unknown-time";

    if (time_info != NULL) {
        if (strftime(timestamp, sizeof(timestamp),
            "%Y-%m-%d %H:%M:%S", time_info) == 0) {
            snprintf(timestamp, sizeof(timestamp), "unknown-time");
        }
    }

    fprintf(file, "[%s] ERROR: %s\n", timestamp, message);
    fclose(file);

    [ Read 27 lines ]
    ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^-A Set Mark
    ^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_/ Go To Line ^-E Redo      ^-G Copy
```

```
GNU nano 7.2 calculator.c
#include <string.h>
#include "calculator.h"

int calculate(const char *operation, double a, double b, double *result)
{
    if (operation == NULL || result == NULL)
        return 0;

    if (strcmp(operation, "add") == 0) {
        *result = a + b;
    }
    else if (strcmp(operation, "subtract") == 0) {
        *result = a - b;
    }
    else if (strcmp(operation, "multiply") == 0) {
        *result = a * b;
    }
    else if (strcmp(operation, "divide") == 0) {
        if (b == 0.0)
            return 0;
        *result = a / b;
    }
    else {
        return 0;
    }
}
```

```
GNU nano 7.2 main.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <errno.h>
#include <math.h>

#include "calculator.h"
#include "logger.h"

static int parse_number(const char *text, double *number)
{
    char *end;
    errno = 0;

    double value = strtod(text, &end);

    if (text == end || *end != '\0' ||
        errno == ERANGE || !isfinite(value)) {
        return 0;
    }

    *number = value;
    return 1;
}

int main(int argc, char *argv[])
{
    [ Read 64 lines ]
    ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
    ^C Location  M-U Undo     M-A Set Mark
    ^_/ Go To Line M-E Redo     M-6 Copy
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ gcc -Wall -Wextra -std=c11 main.c calculator.c logger.c -o skill21 -lm
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ls
calculator.c calculator.h hi logger.c logger.h main.c skill21
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ |
```

Run the successful test cases:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 add 10 5
Operation: add
Input values: 10.00 and 5.00
Result: 15.00
Execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 subtract 10 5
Operation: subtract
Input values: 10.00 and 5.00
Result: 5.00
Execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 multiply 4 3
Operation: multiply
Input values: 4.00 and 3.00
Result: 12.00
Execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 divide 20 4
Operation: divide
Input values: 20.00 and 4.00
Result: 5.00
Execution completed successfully.
```

Test runtime errors and recovery:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 divide 10 0
Error: Division by zero is not allowed.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 power 2 3
Error: Unsupported operation 'power'.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 add abc 5
Error: Invalid numeric input.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 add 10
Usage: ./skill21 <add|subtract|multiply|divide> <number1> <number2>
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21
Usage: ./skill21 <add|subtract|multiply|divide> <number1> <number2>
```

Verify the error log:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ cat error.log
[2026-10-10 05:45:07] ERROR: Division by zero attempted.
[2026-10-10 05:45:13] ERROR: Unsupported calculator operation.
[2026-10-10 05:45:17] ERROR: Invalid numeric input detected.
[2026-10-10 05:45:21] ERROR: Invalid command syntax or incorrect number of arguments.
[2026-10-10 05:45:28] ERROR: Invalid command syntax or incorrect number of arguments.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 divide 10 0
Error: Division by zero is not allowed.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ^C
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 add 7 3
Operation: add
Input values: 7.00 and 3.00
Result: 10.00
Execution completed successfully.

```

Final project verification:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ls -l
total 48
-rw-r--r-- 1 bhardwaj bhardwaj 597 Oct 10 05:40 calculator.c
-rw-r--r-- 1 bhardwaj bhardwaj 125 Oct 10 05:38 calculator.h
-rw-r--r-- 1 bhardwaj bhardwaj 410 Oct 10 05:53 error.log
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 7 14:08 hi
-rw-r--r-- 1 bhardwaj bhardwaj 650 Oct 10 05:39 logger.c
-rw-r--r-- 1 bhardwaj bhardwaj 80 Oct 10 05:39 logger.h
-rw-r--r-- 1 bhardwaj bhardwaj 1595 Oct 10 05:41 main.c
-rwxr-xr-x 1 bhardwaj bhardwaj 16768 Oct 10 05:41 skill21

```

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ gcc -Wall -Wextra -std=c11 main.c calculator.c logger.c -o skill21 -lm
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$ ./skill21 add 25 15
Operation: add
Input values: 25.00 and 15.00
Result: 40.00
Execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-21$

```